

Critical Design Report

Automated Refueling Cislunar Outpost (ARCO)

Team 6

3/21/2023

Authors:

Alexander Uribe
BS-Aerospace Engineering

Summer Huang
BS-Aerospace Engineering

Ernest John Franco
BS-Aerospace Engineering

Andrew Jeffrey Mills
BS-Aerospace Engineering

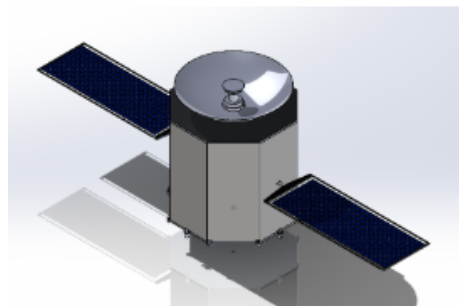
Angelica Cortez
BS-Aerospace Engineering
BA-Physics

Samuel Estrada
BS-Aerospace Engineering

Elizabeth Cotto
BS-Aerospace Engineering

Juan Flores
BS-Aerospace Engineering

Andrea Coca Gonzales
BS-Aerospace Engineering
Minor-Geographic
Information Science



Executive summary

Automated Refueling Cislunar Outpost (ARCO) will be launched into a LEO parking orbit by the Falcon Heavy launch vehicle. From this parking orbit the Falcon Heavy will perform a trans-L1 injection, sending ARCO on a trajectory to the Earth-Moon L1 point. Upon reaching L1, Phase 1 will begin and the L1 orbit insertion maneuver will occur. After the insertion, ARCO will be released from the launch vehicle. In Phase 2, commissioning occurs. In this phase, ARCO will commence deployment of its components and perform all necessary tests to ensure everything is in working condition after launch. At Phase 3, servicing begins and our platform will refuel approximately 50 mid-size spacecraft with 225 kg of Hydrazine each for a total of 11,250 kg of Hydrazine over the 10-year mission life. ARCO will be continually pointing its high gain antenna at the Earth, and solar panels will be articulated to provide constant power to the solar panels. The orbit will have to be maintained by the SCAT thrusters with a low delta-v maneuver approximately every 21 days. Reaction wheel momentum will have to be offloaded by the MRE thrusters with low delta-v burns approximately every 23 days. During Phase 4 (which will run concurrently with Phase 3) ARCO will perform radiation assessments of the L1 environment. At the end of mission life, phase 5 will begin, at which point our platform will cease orbit maintenance, falling into a high Earth orbit.

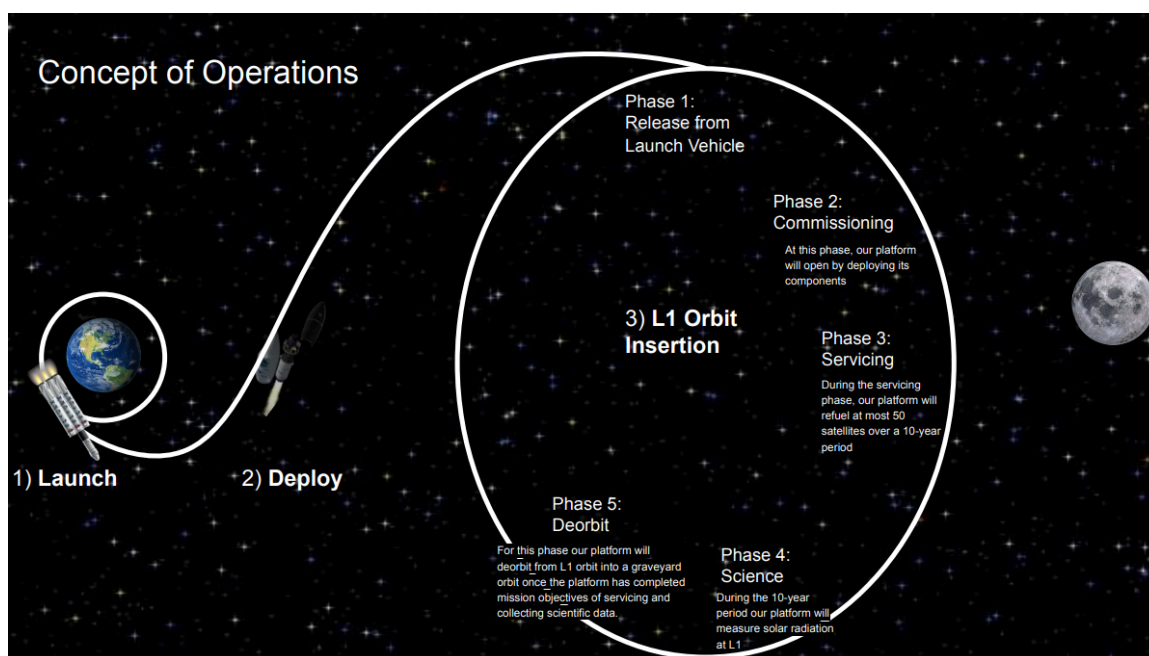


Table of Contents

Executive summary	1
1. Introduction	5
1.1. Project Scope	5
1.2. Customer/System Requirements	5
1.3. System overview	5
2. Applicable documents	6
2.1. Government documents	6
2.2. Non-government documents	6
3. System	6
3.1. Definitions	6
3.2. Characteristics	6
3.2.1. Performance	6
3.2.2. External interface characteristics	7
3.2.3. Physical characteristics	7
3.2.4. Other characteristics	7
3.3. Design	8
3.3.2. Analysis tools	8
3.3.3. Trade studies and selection	8
3.4. Integration	8
3.5. Validation/Qualification	8
3.6. Cost	9
4. Propulsion Subsystem	10
4.1. Definitions	10
4.2. Characteristics	10
4.3. Design	12
4.4. Construction	12
4.5. Verification	12
5. GNC Subsystem	13
5.1. Definitions	13
5.2. Characteristics	13
5.2.1. Performance	13
5.2.3. Physical characteristics	14
5.2.4. Other characteristics	14
5.3. Design	14
5.3.1. Design approach	14
5.3.2. Analysis tools	15
5.3.3. Trade Studies	15

5.4. Construction	16
5.5. Verification	17
6. Avionics Subsystem	18
6.1. Definitions	18
6.2. Characteristics	18
6.2.1. Performance	18
6.2.2. External interface characteristics	18
6.2.3. Physical characteristics	18
6.2.4. Other characteristics	19
6.3. Design	19
6.3.1. Design Approach	19
6.3.2. Analysis Tools	19
6.3.3. Trade Studies	19
6.4. Construction	21
6.5. Verification	23
6.6. Science	24
6.6.1. Characteristics	24
6.6.1.1. Performance	24
6.6.1.2. Physical characteristics	24
6.6.2. Design	25
6.6.2.1. Design approach	25
6.6.2.2. Analysis tools	25
6.7. Communication	25
6.7.1. Characteristics	25
6.7.1.1. Performance	25
6.7.1.2. Physical characteristics	25
6.7.2. Design	26
6.7.2.1. Design approach	26
6.7.2.2. Analysis tools	26
6.8. Power	26
6.8.1. Characteristics	26
6.8.1.1. Performance	26
6.8.1.2. Physical characteristics	26
6.8.2. Design	27
6.8.2.1. Design approach	27
6.8.2.2. Analysis tools	27
7. Structures Subsystem	27
8. Docking Subsystem	33
8.1. Characteristics	33
8.1.1. Performance	33

8.1.2. External Interface Characteristics	33
8.1.3. Physical Characteristics	33
8.1.4. Other Characteristics	33
8.2. Design	34
8.2.1. Design Approach	34
8.2.2. Analysis Tool	34
8.2.3. Trade Studies and Selection	34
8.3. Construction	35
8.4. Verification	35
9. Appendices	36
A. Individual contribution	36

1. Introduction

1.1. Project Scope

Our Automated Refueling Cislunar Outpost (ARCO) will be capable of being launched into the Earth-Moon Lagrange point 1 (L1) in a single launch. Our platform will then maintain in the L1 orbit for at least 10 years. During those 10 years of orbit at L1 our platform will be able to refuel customer spacecraft with Hydrazine propellant. By refueling other cislunar spacecraft it will help reduce future cislunar spacecraft launches by increasing their lifespan. During the fueling process, our platform will be captured by the customer satellite, refuel them, and be released when refueling is finished. ARCO will collect scientific data by sensing the solar radiation at L1 over a 10-year period, helping NASA to furthering knowledge of the L1 environment.

1.2. Customer/System Requirements

The customer requirements for this platform were derived from (RASC-AL). These requirements address the need for additional services and logistics for cislunar space. The table below lists the requirements that must be met for a successful mission. System requirements will be discussed in each section of the report according to subsystems.

Table 1.1.: Customer Requirements

REQ ID	Requirement
CR1.0	L1-Platform shall be at LaGrange point 1
CR2.0	L1-Platform shall be launched in a single launch
CR3.0	L1-Platform shall have a lifecycle of at least 10 years
CR4.0	L1-Platform shall be launched no later than 2030
CR5.0	L1-Platform shall offer in-orbit services to other spacecraft
CR6.0	L1-Platform shall have the ability to conduct and collect scientific data
CR7.0	L1-Platform mission shall be completed on an economic budget
CR8.0	L1-Platform shall be reliable
CR9.0	L1-Platform shall be easily maintained

1.3. System overview

If everything goes right during a spacecraft mission, end-of-life comes with the depletion of propellant. With the rapidly increasing tempo of launches past Earth's orbit, there is a growing need to provide fuel to these spacecraft farther away than low Earth orbit. To this end, we have designed the Automated Refueling Cislunar Outpost (ARCO). ARCO will be transported into the Earth-Moon Lagrange point 1 (L1) in a single launch and will be capable of refueling approximately 50 midsize customer spacecraft. Our platform will maintain orbit in L1 for at least 10 years. During those 10 years of orbit at L1, ARCO will be able to provide other cislunar spacecraft with much-needed Hydrazine propellant. Customer satellites will dock with the ARCO spacecraft, refuel over a short period, and release themselves. By refueling cislunar spacecraft, their mission range can be drastically increased, allowing for explorations further through our solar system than previously possible. ARCO also contains an onboard spectrometer which will take data of the L1 environment, furthering decadal science goals and providing long-term data for future L1 operations. ARCO was designed using space-proven off-the-shelf components, so

all components on board have a TRL of 7 or above. ARCO will greatly improve upon the state of the art by allowing spacecraft to be refueled with Hydrazine at orbits much farther from Earth than previously possible. By utilizing an innovative structure and a refueling-focused design, ARCO will be capable of carrying 11,250 kg of Hydrazine propellant, enabling larger-scale missions than any refueling mission currently available to use our services.

2. Applicable documents

2.1. Government documents

Our System must first adhere to the *Evolved Expendable Launch Vehicle Rideshare User's Guide* [2.1] in order to be compatible with the EELV Standard Interface. Additionally, our systems must follow standards laid out by NASA and the NASA Technical Standards System.[2.2]

2.2. Non-government documents

ARCO's design relies on the user guides of each of our off-the-shelf components, and has referenced *Space Mission engineering: The New SMAD* [2.3] for examples of calculations used throughout our design.

3. System

3.1. Definitions

Solar Array - assembly of solar panel design featuring stowed and deployed configurations implementing a roll out mechanic

High-Gain Antenna - device used to communicate and receive information to and from Earth to L1

Rafti - universal docking port used to connect to and refuel other spacecrafts

Spectrometer - device used to take radiation readings from the sun

Star Trackers - devices used to determine location and navigate through L1 orbit

Reaction wheels - Device used to reorient spacecraft in combination with the propulsion system in order to maintain orbit in L1 every 23 days

SCAT/MRE1 thrusters - propulsion system used to make small and large maneuvers depending on requirement

Whipple Shield - structural shielding used to protect spacecraft vitals from solar radiation and miniature space debris

3.2. Characteristics

3.2.1. Performance

ARCO will be launched in a single flight to L1 with a total wet mass of 16,649.595 kg. Included in the wet mass will be 12500 L (11,250 kg) of customer- reserved Hydrazine, which will be dispensed to docked customers at a rate of 4 L/min (0.0667 kg/s). ARCO's bipropellant stationkeeping thrusters will provide 50.9 m/s of Delta-V for stationkeeping to the L1 orbit. ARCO will have a peak power consumption of 1,039.61 W, and use 90,012.38 kJ of energy at peak. Reaction wheels will maintain a pointing accuracy of 5 arcseconds throughout operation. The High-Gain antenna will transfer data at a rate of 4.25 Gbps. ARCO's Spectrometer will take readings in the $10^{(-10)}$ to $10^{(-12)}$ nm range.

3.2.2. External interface characteristics

Customer satellites will need a RAFTI compatible docking claw [8.2] and RAFTI refueling port [8.2] in order to receive fuel from our station. Ground stations will need to be able to send and receive data over the Ku bandwidth of 12 to 18 GHz and Ka bandwidth of 27 to 40 GHz and receive data at a rate of 4.25 Gbps. ARCO will be attached to the Falcon Heavy launch vehicle using an EELV [2.1] compliant payload adapter.

3.2.3. Physical characteristics

Figure 3.1: 3-D Drawing of Arco

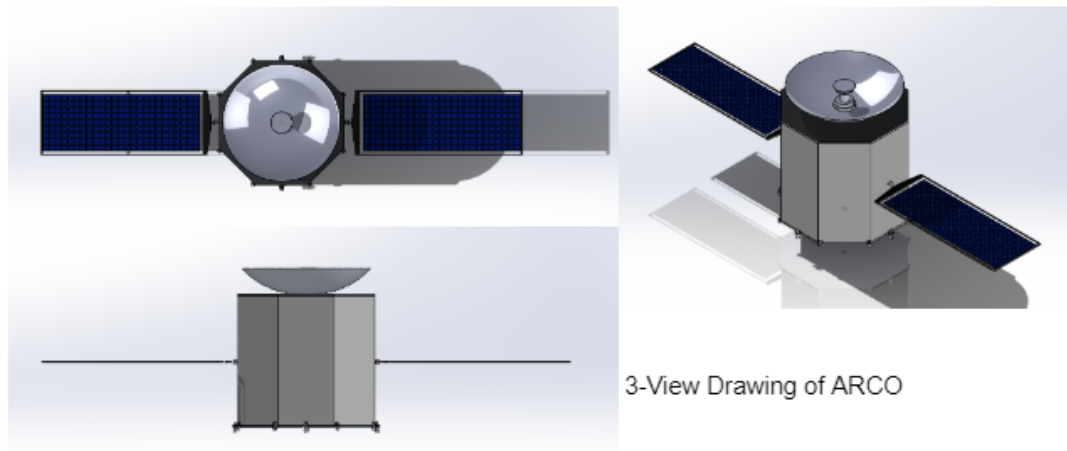


Table 3.1. System mass distribution

Mass (kg)	CDR (kg)	PDR (kg)	SRR (kg)
Structures	4,779.939	4,175.6	n/a
Docking	0.550	8,272.55	n/a
Propulsion	11,542.690	3,327.89	3,900
Avionics	252.95	23.275	5.7
GNC	7.185	31.81	n/a
Total	16,583.314	16,063.875	<16,800

3.2.4. Other characteristics

Design features a stowed and deployed configuration in order to satisfy the launch requirement of fitting into the Falcon Heavy fairing. The structure is wrapped by a layer of kevlar shielding that has the capability of resisting the radiation exposure from the sun and deflecting miniature debris located in L1 orbit.

3.3. Design

At the system level, the major parameter that was used to finalize the integrated system design was mass. Since PDR, the total dry mass of the structure has been shaved by over 10,000 kg by shortening the height of the spacecraft and the diameter of the frame, while maintaining the

same avionics components of the spacecraft. Also, the solar array design was adjusted to account for the requirement that the star trackers must not be facing the sun at all times.

3.3.1. Design Approach

The design approach used to develop the spacecraft originally had a vision of using the maximum amount of allowable volume inside of the fairing of the Falcon Heavy launch vehicle. The major change from PDR to CDR involved the discovery that the mass requirement was a severe limiting factor that prevented the spacecraft from carrying an abundant amount of refueling propellant. To achieve this, three things were done to make sure the design followed requirements: finding the required power estimate of the entire spacecraft in order to properly size the solar array, finding the spherical volume of the propellant tanks using the density of 12,500 L of hydrazine and converting, and sizing a structural frame to hold all components and withstand the load force and axial force of launch.

3.3.2. Analysis tools

Analysis was performed from the baseline assumption that all component user guides and references were accurate. Tools used for analysis were: Solidworks, Autodesk Inventor, MATLAB, and Simulink. Analysis performed using Solidworks and Autodesk Inventor used accepted values as inputs that came from the database of those two softwares. Outputs included mass estimations, displacements under axial loading, and stress concentrations under axial loading. These analyses informed design decisions. MATLAB and Simulink took inputs of spacecraft moment of inertia and center of gravity location and outputted the size of the reaction wheels and estimated momentum buildup.

3.3.3. Trade studies and selection

Solar Panel Trade Study: Researched concept of “Roll Out Solar Array” design in order to implement features of solar panels that can reorient to constantly face directly towards the sun. Whipple Shielding: Researched concept of whipple shield layer to reinforce structural frame and protect spacecraft vitals from debris and radiation

3.4. Integration

Integration of the system involves first constructing the frame of the spacecraft by welding a series of I-beams together, followed by mounting the spherical propellant tank to the center ring of welded beams. The whipple shield and all other avionic components are fastened outside of the frame.

3.5. Validation/Qualification

Table 3.2. Validation/Validation Tool

Orbital Insertion	STK simulation of the maneuver with a delta-v and fuel use calculation
Orbital Maintenance	STK simulation of the orbit with delta v and fuel use totals over 10-yr span
Reaction Wheel Momentum Dump	MATLAB simulation of accrued angular velocity and torque required to dissipate.
Moment of Inertia	Solidworks evaluation of the fully integrated system will provide the moment of inertia.

Launch Axial Load	Solidworks simulation of the fully integrated system under launch load.
Launch Vibration (Random and Sinusoidal)	Solidworks simulation of the fully integrated system under random and sinusoidal vibration loads.
Mass	Solidworks evaluation of the fully integrated system will provide the total mass.

3.6. Cost

Table 3.3.: Component Costs

Component	Quantity/Run	Total Quantity	Price/Unit (\$)	Margin	Total Price (\$)
star trackers	2	6	140,000	28,000	868,000
reaction wheels	3	9	75,000	15,000	690,000
Computer (OBC)	1	3	1,500,000	300,000	4,800,000
antenna/twta	1	3	65,000	13,000	208,000
radio	1	3	100,000	20,000	320,000
spectrometer	1	3	1,000,000	200,000	3,200,000
data acquisition system	1	3	~60,000	12,000	192,000
solar panels	2	6	~20,000	4,000	124,000
nickel hydrogen batteries	56,940 (hr)	60,000 (hr)	100/kW-hr	2,000	6,000,000
rafti dock	1	1	45,000	9,000	54,000
propellant				8 million	48,000,000
heaters	1	1	~400,000	800,000	1,200,000
structure	1	1	~65,000	13,000	78,000
construction/facilities	n/a	n/a	n/a	n/a	20,000,000
Total without Launch					85,734,000
Falcon Heavy (Launch Vehicle)	1	1	90 mil		90,000,000
Total w/ Launch					175,734,000

The cost budget includes materials for prototype, testing, and final launch. Certain instruments took the three stages into account, which increased the cost of the mission. All major components such as RAFTI dock, propellant, heater and structure were all assumed to have one for the mission, as they will only be used while at Lunar LaGrange Point 1.

Table 3.4.: Income

Benefactor	Amount (\$)
Customer	15,000,000
Total	750,000,000

The income chart includes the cost of refueling per customer, plus all funding that will be needed in order to manufacture. The customer cost is valued to be 15 million for 225kg of fuel per customer. Expected number of customers to be refueled by ARCO is about 50. Therefore, our total income is 15 million*50 customers which gives us our total of 750 million.

4. Propulsion Subsystem

4.1. Definitions

Throughput: mass flow rate

Specific Impulse: the total impulse divided by the weight of the propellant

Wet Mass: the total mass of the vehicle

Dry Mass: the rest of the mass after all propellant is used

Delta-V: change in velocity of a spacecraft after propellant is fired

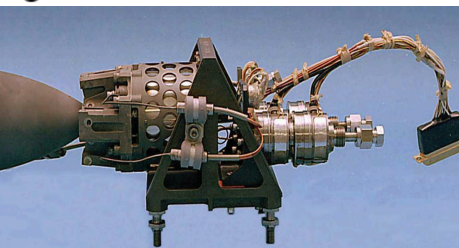
4.2. Characteristics

To make sure ARCO has the ability to stay in Earth-Moon LaGrange point 1 for 10 years while refueling for customer satellites, and collecting scientific data, Northrop Grunman's thrusters: Secondary Combustion Augmented Thruster TR-500 (SCAT TR-500) and MRE-1.0 are used for the mission. ^[4.1]

4.2.1. Performance

4.2.1.1. TR-500

Figure 4.1 SCAT TR-500 Thruster



It has two modes of combustion: monopropellant or bipropellant. It can use the fuel more efficiently and deplete the full tank with nothing unused fuel left. Its deep blowdown and thrust modulation features enhance the mission flexibility and effectiveness. Its efficient bipropellant mode uses dinitrogen tetroxide and hydrazine. Its regenerative-cooled oxidizer chamber gives it a better cooling margin and makes it insensitive to mixture ratio swings. Following is a summary table of its performance.

Table 4.1. SCAT TR-500 Characteristics

Thrust Biprop Mode	17.8-62.3 N	Mixture ratio	0.95 to 1.6	Inlet Pressures	90 to 600 psia
Thrust Modulation	30% delta max	Specific Impulse	325s max	Qualification Life	90,000 seconds

4.2.1.2. MRE-1.0

Figure 4.2 MRE-1.0 Thruster



The MRE-1.0 will be used to unload momentum accumulated in the reaction wheels and used for attitude control when reaction wheels are reset. Following is a table of its performance summary.

Table 4.2. MRE-1.0 Characteristics

Thrust at maximum operating pressure	5.0 N at 400 psia	Steady state specific impulse at 275 psia inlet pressure	218 sec	Maximum throughput	544 kg
Thrust at 275 psia inlet pressure	3.4 N	Operating pressure range	8-565 psia	Maximum cycles	457,849

4.2.2. External Interface Characteristics

The thrusters will interface with a fuel tank only for fuel moving ARCO itself such as station keeping, momentum unloading and attitude control. The fuel tank chosen is Ariane Group's Bipropellant Tank model OST 25/3 with a net volume of 331 liters.^[4.2]

4.2.3. Physical Characteristics

Table 4.3. Physical Characteristics of TR-500 and MRE-1.0

Thruster	MRE-1.0	TR-500
Weight	0.5 kg	2.2kg
Envelope (width x length in mm)	114 x 188	312.42 x 114.3 x 139.7

4.2.4. Other Characteristics

Solenoid pressure valves are used to control liquid fuel propellant mass flow rate.

Following is a table of summary of Moog bipropellant thruster valve.^[4.3]

Table 4.4. Summary of Moog bipropellant thruster valve.

Max Operating Pressure, MEOP [psia(bar)]	275 (19)	Burst Pressure [psia(bar)]	1200 (82.8)	Operating Voltage Range [Vdc]	49 to 51	Close Response Time [msec]	6 max
Proof Pressure [psia(bar)]	765 (52.8)	Flow Coefficient [GPM water/(psid) ^{0.5}]	0.00043	Open Response Time [msec]	6 max	Power Consumption [watts]	217 at 51 Vdc, 70F
Leakage per Seat, Internal [scc/hr]	3	Cycle Life [cycles]	1,000,000	Inlet Filtration [micron absolute rating]	25	Representative Model Numbers	-051-178-5
Leakage, External [scc/s]	1.00E-06	Weight [lbm (gram)]	0.4 (182)	Operating Temperature Range[°F (°C)]	40 to 250 (4.4 to 121)		

4.3. Design

4.3.1. Design Approach

Each step of design was to meet the customer requirements while with a possible clear verification in mind. As the propulsion subsystem is to support the refueling and data collection, highest TRL is also one of the top considerations for design. All thrusters, valves and tank are flight proven with a technology readiness level of 9.

4.3.2. Analysis Tools

Solidworks was used to construct models of the thrusters and to find the center of mass and best position to place the thrusters. Solidworks was also used to size the dimension for the propellant tank for ARCO itself. Research in STK with NASA's documents and patents were used to analyze the fuel usage of Earth-moon LaGrange point 1 station keeping for 10 years. The station-keeping delta-v budget was found to be around 50.9m/s for the total 10-year lifespan duration.

4.3.3. Trade Studies

Table 4.5. Propulsion System Trade Studies

Components	Mass/ Dimensions	Thrust	Specific Impulse	TRL
Northrop Grumman TR-500 SCAT	2.2 kg/12.3" x 4.5" x 5.5"	17.8-62.3 N	325s max	9
Northrop Grumman MRE-1	0.5 kg/114mm x 188mm	5.0 N at 400 psia	218 sec	9
Ariane Group 10N Bipropellant Thruster	350 g	10N	292 s	9
Ariane Group 1N Monopropellant Thruster	290 g	0.320 to 1.1 N	220 s	9

4.4. Construction

Part list: Northrop Thruster SCAT TR-500, MRE-1.0, Moog bipropellant thruster valves, monopropellant thruster valves, Ariane Group bipropellant tank.

Manufacturing: The thrusters come with built-in truss that connects to the outer structure fixture. Fuel lines and fittings will be used to connect the thrusters, valves and fuel tank.

4.5. Verification

Rocket equation was used to verify the estimated fuel weight of station-keeping for 10 years of the ARCO lifespan.

Equation 4.1 Rocket Equation

$$\Delta v = v_e \ln \frac{m_0}{m_f} = I_{sp} g_0 \ln \frac{m_0}{m_f},$$

NASA's research papers and research documents in STK were used for verifying fuel usage for station keeping.^{[4.4][4.5][4.6]}

5. GNC Subsystem

5.1. Definitions

- 1) Three-axis configuration: Configuration where reaction wheels are located in the x,y, and z planes to facilitate yaw, pitch, and roll.
- 2) Center of mass: The center of mass is the relative position in our spacecraft where the mass is balanced. This is the ideal location for reaction wheels.
- 3) Attitude error: The difference between true and measured distances of celestial bodies in space.

5.2. Characteristics

5.2.1. Performance

The GNC subsystem will allow us to know the exact location of our platform in space and rotate the platform to keep its antenna pointed towards Earth. We will be using star trackers, to achieve our requirements of knowing our exact location in space with an attitude error of less than 5 seconds. We will also be using reaction wheels to generate an angular momentum of $33600 \text{ kg} \frac{\text{m}^2}{\text{s}}$ that will be sufficient to rotate our spacecraft.

5.2.1.1. Reaction Wheels

The reaction wheels that were chosen were the Rockwell Collins wheels. These wheels were chosen due to their ability to maneuver 5000 kg per wheel. The wheel has a diameter of 347 mm and a height of 124 mm^[5.1].

5.2.1.2. Star Trackers

The two star trackers in our platform are the ST-16RT2 trackers manufactured by RocketLab. They will be mounted onto the solar panels and will be recalibrated every time the panels move. Each tracker has a dimension of 98.81 mm x 119.5 mm x 54.4 mm. The attitude error of the trackers is 5 arcseconds^[5.2].

5.2.2. External interface characteristics

5.2.2.1. Reaction wheels

The wheels do not have external interfaces.

5.2.2.2. Star Trackers

The star trackers will measure the angle between nearby stars and use those measurements to calculate their exact position in space. The trackers are also equipped with a system that allows them to recalibrate in space^[5.3].

5.2.3. Physical characteristics

5.2.3.1. Reaction Wheels

The reaction wheels have a height of 124 mm and a diameter of 347 mm. Each reaction wheel will have a mass of 6.3 kg.

5.2.3.2. Star Trackers

The star tracker is 98.81 mm long and 54.4 mm wide. The base of the tracker measures 23.5 mm and a large baffle measures 95 mm. Adding the length of both components gives us the total height of the star tracker which is 119.5 mm. Each star tracker weighs 230 grams.

Table 5.1: GNC COMPONENT PHYSICAL CHARACTERISTICS

COMPONENT	MASS (kg)	DIMENSIONS (mm)	QUANTITY
Reaction Wheels	6.3	347 (diam) x 124 (height)	3
Star Trackers	0.23	95 (h) x 119.5 (l) x 98.81 (w)	2

5.2.4. Other characteristics

5.2.4.1. Reaction Wheels

The reaction wheel will also include an enclosure this enclosure is .421 m x .421 m x .147 m. This enclosure will ensure the reaction wheels are mounted safely.

5.2.4.2. Star trackers

The star trackers also have a survival temperature that ranges from -40°C to $+95^{\circ}\text{C}$. The lens is a full-custom 16 mm f/1.6 4-element glass designed for shock and vacuum. The field of view measures $7.5^{\circ} \times 10^{\circ}$ half-angle ^[5.2].

5.3. Design

5.3.1. Design approach

5.3.1.1. Reaction Wheels

The reaction wheels are off-the-shelf components obtained from Rockwell Collins. The model RSI 18-220/45 was chosen because each wheel generates a torque of about 220 mNm which is sufficient to rotate our spacecraft.

5.3.1.2. Star Trackers

The star trackers are off-the-shelf components obtained from RocketLab. The model ST-16RT2 was chosen because its attitude error complies with our subsystem requirement.

5.3.2. Analysis tools

5.3.2.1. Reaction wheels

MATLAB was used to perform calculations and figure out the right performance characteristics for our reaction wheels. Using MATLAB allowed us to easily update the reaction wheels' angular momentum required as our mission and spacecraft characteristics changed. The verification process is discussed in more detail later in the section.

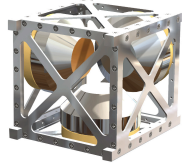
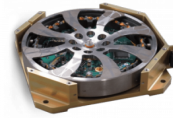
5.3.2.2. Star Trackers

No analysis is needed to make sure the star trackers meet the initial 5 arcsec error requirement since it is a characteristic of the chosen component. However, after deciding to place the tracker on the panels, Simulink was used to verify that the panel motion would not increase our attitude error. The verification process is discussed in more detail later in the section.

5.3.3. Trade Studies

5.3.3.1. Reaction Wheels

Initially, the reaction wheels were going to be placed in a box (as seen in figure 1) at the platform's center of mass. This design would facilitate the mounting of the wheels and ensure the origin of the three-axis configuration would be at the exact center of mass ^[5.4]. However, this location would interfere with the propellant tanks and it would also require some thermal isolation since the storage temperature range of the propellant and the reaction wheels' operational temperature range are different. After performing a trade study, it was decided to move the wheels to the outside of the platform and mount them and protect them with an enclosure (refer to figure 2) to give the propellant tanks more space and to fix the operational temperature issue.

Figure 5.1: Initial mounting design selected**Figure 5.2: Mounting design****5.3.3.2. Star Trackers**

The trade studies performed to select the right component were based on comparing the attitude error, power consumption, weight, and price of multiple off-the-shelf star trackers.

Table 5.2: Star Trackers Trade Study

Model	Vendor	Attitude Error	Size (mm)	Mass (kg)	Power (W)	Update Rate (Hz)	Cost (USD)
ST200	Hyperion Technologies	< 30 arcsec	29 x 29 x 38.1	.42	6	5	1,000 – 10,000
ST-26RT2	Rocket Lab	5 arcsec	119.5 x 95 x 98.81	.23 k	<1	2	140,000

5.4. Construction**5.4.1. Reaction Wheels****5.4.1.1. Manufacturing**

The reaction wheels will be bought from Rockwell Collins. This company specializes in space-grade reaction wheels. In order to mount our reaction wheels onto our spacecraft we will be required to make a mounting bracket that will be made of the same material our structure is made from.

5.4.1.2. Additional Information

The mounting brackets for our reaction wheels will be made of 6061 aluminum alloy. This will be milled out of the same material our structure is made of. The design can be easily replicated on a CNC machine or cast due to its simple design. The cost of this will be \$415.44 for 3 mounting brackets. This will include the cover and base.

5.4.2. Star Trackers

5.4.2.1. Manufacturing

- 5.4.2.1.1. The two ST-16RT2 star trackers purchased from RocketLab will be mounted onto the solar panels using their triangular M3 screw arrangement and dowel pins. Washers or screw heads with an 8 mm diameter are ideal for this type of screw arrangement. The star trackers' wires will be placed inside the shallow solar panel frame.

5.4.2.1.2 Additional Information

The bodies of the star trackers are made of aluminum and the rest of the components are black anodized. The bottoms of the three mounting feet have yellow chemical conversion coating which is electrically conductive and is used as the grounding path for the device. The cost of each tracker is \$140,000^[5.2].

5.5. Verification

5.5.1. Reaction Wheels

MATLAB was used to verify the math and make sure the reaction wheels produced enough angular momentum to rotate the platform. First, disturbances in space were calculated, such as solar torque and momentum gained after one orbit. After that, our spacecraft's moment of inertia was calculated and the results obtained helped us determine the characteristics of the reaction wheels needed to meet our initial requirements. Figure 5.3 and Figure 5.4.

Figure 5.3: Reaction wheel sizing

```
format long g
%Solar Torque
I0 = 1370; %w/,^2
c = 2.998*10^8;%m/s
Ps = I0/c;
As = 5.03*2.51;
Lc = 326400*1000;
Fs = (1+.4)*(Ps)*(As)*(cosd(54.7));
Ts = 2*Fs*(2.515+2.3);

M = Ts*39000*60;% momentum build up over 1 orbit

x = 4;
Rx = [1 0 0; 0 cosd(x) -sind(x); 0 sind(x) cosd(x)];
Ry = [cosd(x) 0 sind(x); 0 1 0; -sind(x) 0 cosd(x)];
Rz = [cosd(x) -sind(x) 0; sind(x) cosd(x) 0; 0 0 1];

R = Rz*Ry*Rx;

MOI = .5*16800*2^2; %moment of inertia of spacecraft
```

Figure 5.4: Reaction wheel sizing output

```
The momentum build up over 1 orbit is 1052 kgm^2/s
The solar torque is 0.0004494753
Our MOI is 33600
Our spacecraft mass is 16800 kg
```

5.5.2. Star Trackers

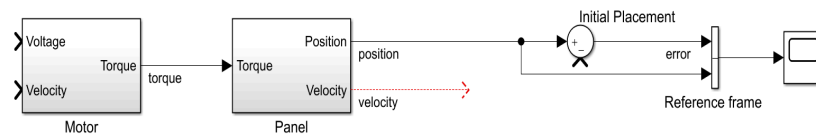
Simulink was used to model the star trackers' recalibration when the solar panels move. The new position of the panel can be used using the following equation:

Equation #5.1: Panel equation of motion

$$\frac{d^2\theta}{dt^2} = \frac{1}{J} \left(T - K_d \frac{d\theta}{dt} \right).$$

After receiving the panels' new position as input, the star trackers will collect new angle of separation data from its selected celestial bodies, compare the data to the old data stored, and compute a new reference frame.

Figure 5.5: Star Tracker Recalibration



6. Avionics Subsystem

6.1. Definitions

- 1) Power: $P = \frac{\text{energy}}{\text{time}} = \frac{E}{T}$; defines the power consumption of ARCO and the power needed to be produced to power all instruments over the 10 year project
- 2) Energy: $E = \text{power} \cdot \text{time} = P \cdot T$; a variation of the power definition where energy is divided into power of each component with regards to time it must be on.
- 3) Spectrometer: a device that detects radiation in space
- 4) Traveling Wave Tube Amplifier (TWT): amplifies an electromagnetic wave in order to transmit data

6.2. Characteristics

6.2.1. Performance

The total mass of the platform is **16, 583.314** kg, and the avionics subsystem contributes about 252.95 kg to that amount. Avionics makes up three parts: science, communications and power. It was found that the total power consumed is 1504.778 W. Power and energy details are calculated in the verification section (6.5)

6.2.2. External interface characteristics

The platform will be sending data to Earth as it collects information from customers and components on board. It will also have to communicate with other satellites in space in order to send its location and provide its planned services.

6.2.3. Physical characteristics

All component drawings/models will be presented in the corresponding sub-subsystem section in order to provide a more detailed explanation.

Table 6.1. Avionics Subsystem mass distribution

Mass (kg)	CDR	PDR	SRR
Solar Panels	200	215	1003.96
Antenna	3	3	9.12
Traveling Wave Tube Amplifier (TWTA)	22.2	0	0
Data Acquisition System	0.05	0.05	0.43
On-Board Computer	5	0	0
JPL/SDL Radio	1.2	1.2	3.64
Spectrometer	1.5	1.5	12.82
Rechargeable Batteries	20	12	56.04
Total	252.95	256.025	1086

6.2.4. Other characteristics

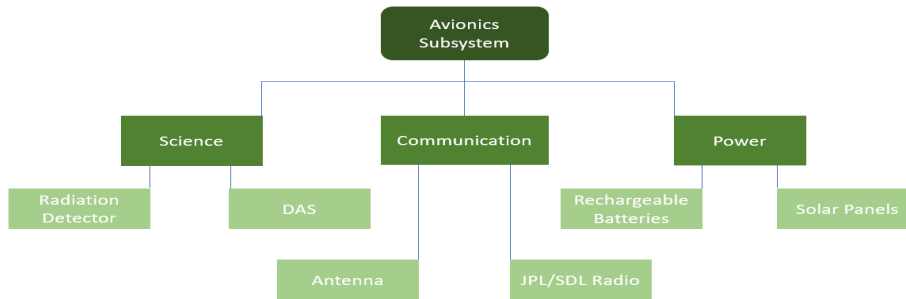
All components for the avionics subsystem were chosen from existing projects, so all have high TRLs. This decision was made to increase the reliability of ARCO and success at the environment of Lunar LaGrange Point 1. All components chosen have shown to be able to within the length of the mission (10 years).

6.3. Design

Compared to PDR no major changes were made to the avionics subsystem overall. The only component that was decided to vary was the type of secondary battery. Originally lithium-ion batteries were chosen, however, with more research it was found that nickel hydrogen is less toxic/explosive. The shift in battery type was made as a safety precaution.

6.3.1. Design Approach

Figure 6.1: Avionics Methodology



6.3.2. Analysis Tools

The analysis done was researching what components could be used for ARCO, based on what the requirements for the platform were and what other companies have used in the past. Upon researching calculations were also done in order to ensure that requirements were met.

6.3.3. Trade Studies

Avionics had two sets of trade studies done. The first was determining how to collect radiation data and which instrument would be the best for characterizing that data we want to observe. The second was comparing our platforms' power usage and whether it would be best to use: primary batteries, secondary batteries, or both.

Table 6.2. Radiation Data Instrument Trade Study

Product	Spectrometer	Dosimeter
Mass/Dimensions	-mass: 1.5 kg -10.3 cm x 12.2 cm x 20.4 cm -power: 4.2 W	-mass: <1 kg -10cm x 10cm x 5cm -Power: 2.5 w
Reliability	-ability to detect charged (protons and heavy ions up to Fe) and neutral particles (neutrons and gamma rays) -specifics: p, He, ions(Li-O), ions(Mg-Fe), neutrons, gamma rays, electrons, and positrons	-measures the energy absorbed by different particles (electrons, protons, gamma rays) -used on Lunar Reconnaissance Orbiter (LRO) since 2009
Cost	-unknown (assuming an average cost)	-low cost
Dose Range	-determine radiation exposure for human travelers on missions to Mars	-used to determine radiation exposure to astronauts while in space
TRL	9	8

After completing the first trade study, we compared the spectrometer and the microdosimeter, it was determined that for the use of our platform, the spectrometer would be the best instrument for radiation detection.

Table 6.3. Source of Power from Batteries Trade Study

	Primary (non-rechargeable)	Secondary (Rechargeable)	Both
Cost	-more costly to reach same operation time -batteries need to be replaced	-cheaper than primary batteries -recharged by solar panels, overall cost will include solar panels and batteries (therefore most expensive than by themselves)	-higher cost as it includes both batteries and solar panels
Reliability	-cannot be recharged -slower response -certain batteries may leak -do not require maintenance	-efficiency decreased ~3000 cycles -higher energy efficiency	-more reliable as it would include multiple power sources
Mass	Ex. -silver zinc: density of 130 W h/kg, life span 30-90 days wet -lithium sulfur dioxide: density of 220 W h/kg, lifespan 10 years wet (with low discharge rate and slow response)	-relatively lighter than primary batteries (50% mass advantage) -solar panels needed for recharging batteries (whole system would be heavier)	-two different types of power sources would increase mass allocated towards avionics
Battery Capacity	-lower rates of discharge -lower storage life	-higher specific energy	-higher battery capacity
Additional Information	-most likely need to have resupply mission	-recently developed, not enough information publicly available	-most common configuration used in the space industry

The second trade study above was used to determine the best source of power from either a primary battery, secondary battery, or using both. Considering cost, reliability, mass, battery capacity, and any other information, we compared each of the three possibilities. Remembering we will also be housing solar panels on our platform; it was concluded that using both primary and secondary batteries would be inefficient for mass usage. Secondary batteries would be best to use as they are rechargeable, lighter and have a higher efficiency than primary batteries.

6.4. Construction

All components for avionics were chosen from pre-existing missions/companies. Cost of instruments was provided by asking industry personnel and listed in Table 6.4. The research done was all the available information that was provided to the general public.

6.4.1. Spectrometer (RAD)

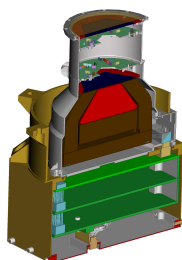
6.4.1.1. Manufacturing

RAD is mounted on the outer shell of ARCO as it needs to be in contact with the environment in order to efficiently collect radiation data. There is only one RAD as this instrument can work on its own. RAD consists of two major pieces described by Hassler as a sensor head (RSH) and electronics box (REB). The RSH houses a solid-state detector telescope for detection, and the REB has three circuit boards for processing signals, communication with ARCO and a power system.

6.4.1.2. Additional Information

Spectrometers work with a data acquisition system (DAS) to store data that is collected and sent back to earth via the antenna. Slice6 Air is the DAS chosen to be compatible with RAD. This device will be stored inside ARCO's body.

Figure 6.2: RAD Detail Cross-section Image



6.4.2. Oxford Space Systems Hinged Rib Antenna

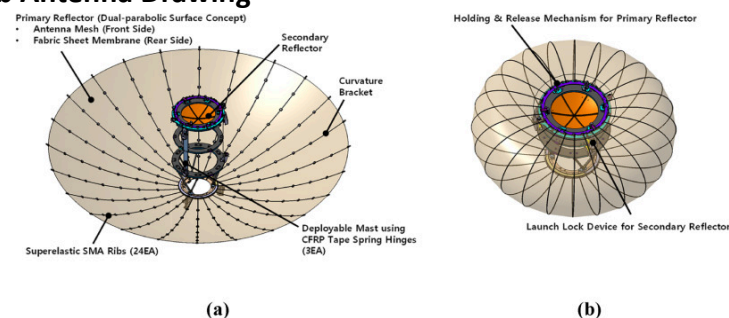
6.4.2.1. Manufacturing

The antenna used for ARCO was chosen off the shelf from Oxford Space Systems. As seen it is mounted on the top of the spacecraft with hinges. All components needed will be provided by Oxford Space Systems, with exception of hinges.

6.4.2.2. Additional Information

Hinges will be a part of the total manufacturing process of ARCO.

Figure 6.3: Hinged Rib Antenna Drawing



6.4.3. JPL/SDL Radio: IRIS V2

6.4.3.1. Manufacturing

Another component vital to sending/receiving data is the JPL/SDL Radio, which is compatible with the chosen antenna. It works together with the Oxford antenna via electrical wiring which will run from the radio to the antenna through the body of ARCO.

6.4.3.2. Additional Information

The JPL/SDL Radio uses the Ka band.

Figure 6.4: Iris V2 Assembly Drawing

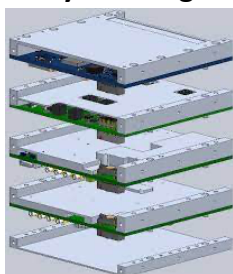


Table 6.4. Cost Budget of Avionics

Component	Quantity/Run	Total Quantity	Price/Unit (\$)	Margin (\$)	Total Price(\$)
Computer (OBC)	1	3	1,500,000	300,000	4,800,000
antenna/twta	1	3	65,000	13,000	208,000
radio	1	3	100,000	20,000	320,000
spectrometer	1	3	1,000,000	200,000	3,200,000
data acquisition system	1	3	~60,000	12,000	192,000
nickel hydrogen batteries	6	60,000 (hr)	100/kW-hr	2,000	6,000,000
Total					14,720,000

6.5. Verification

Two verifications were chosen in order to verify that the Avionics subsystem will meet the subsystem level requirements. The first of these verifications will deal with verifying that the mass of the solar panels are within the range that was predicted during the Preliminary Design Review. The solar panels were estimated to take account for the majority of the mass compared to all other Avionics subsystem components therefore it is important to make sure that the solar panels do not surpass the mass limit for the Avionics subsystem to ensure that the overall mass of ARCO will be achieved. The finalized configuration of the solar panels was decided on through research. The value for mass of the solar panels will be compared to the mass value of the finalized solar panels. If the solar panels are below the value of the predicted mass value then this subsystem requirement is satisfied. For Preliminary Design Review the assumed mass for the panels were 236.5 kg. Now, the finalized design for the solar panels has a mass value of 200 kg. Therefore the solar panels' mass satisfies the requirement. The other requirement that will be verified for the avionics subsystem is to determine whether the energy produced by the

solar panels is enough to power ARCO. For this verification it was decided to calculate the energy produced by the panels in one cycle which will be an earth year of 365 days. In the original calculations for the energy produced by the solar panels no interferences such as eclipse periods were taken into account. Therefore the value of energy produced was compared to the value calculated for energy produced with eclipse periods. In an Earth orbit of about 365 days, Earth-Moon LaGrange point 1 does not see the sun for a time period of 20.7 hours[6.1]. The calculations done for this verification are shown below.

Calculations:

*Recall: Energy = Power * Time. Therefore for all the components:*

$$Energy = P_1 * T_1 + P_3 * T_3 + P_4 * T_4 + P_5 * T_5 + P_6 * T_6 +$$

$$P_7 * T_7 + P_8 * T_8 + P_9 * T_9 + P_{10} * T_{10} + P_{11} * T_{11} + P_{12} * T_{12} \dots\dots\dots (equation 6.1)$$

$$Energy_{total} = \left(16.5 \frac{J}{s}\right)(86,400 s) + \left(3.63 \frac{J}{s}\right)(21,600 s) + \left(2.02 \frac{J}{s}\right)(86,400 s) + \left(297 \frac{J}{s}\right)(86,400 s) + \left(38.5 \frac{J}{s}\right)(43,200 s) \\ + \left(49.5 \frac{J}{s}\right)(86,400 s) + \left(330 \frac{J}{s}\right)(86,400 s) + \left(618.64 \frac{J}{s}\right)(86,400 s) + \left(132 \frac{J}{s}\right)(86,400 s) \\ + \left(4.62 \frac{J}{s}\right)(21,600 s) + \left(1.694 \frac{J}{s}\right)(3632.4 s) \dots\dots\dots (equation 6.2)$$

$$Energy_{Total} = 126,752.5773 KJ$$

$$In\ 365\ days: Energy_{Total(No\ Eclipse)} = (126,752.5773 KJ) * (365\ days) = 46,264,690.71 KJ * day$$

$$Energy_{Total(With\ Eclipse)} = (126,752.5773 KJ) * (364.1375\ days) = 46,155,366.62 KJ * day$$

$$Difference: Estimated - Actual = 46,264,690.71 - 46,155,366.62 = 109,324.09 KJ * day$$

As seen from the calculations the actual energy produced from the solar panels is less than the estimated energy produced which is the ideal case if Earth-Moon LaGrange point 1 had no eclipse periods. Therefore the solar panels did not pass this verification. In order to meet the required energy for ARCO it was decided to add an additional secondary battery to account for the eclipse period. By doing this, the energy would increase to meet our requirement. Another method for meeting the set energy is to add an additional solar panel to our array, increasing area by 10%. This area would keep ARCO's mass within budget. These two validation methods will be further developed once verification is fully run through.

6.6. Science

6.6.1. Characteristics

The science portion will collect radiation at LaGrange Point 1 throughout the mission. The goal of this data collection is to further decadal study by viewing whether humans will be able to reach this specific point, unharmed.

6.6.1.1. Performance

- **Spectrometer: Radiation Assessment Detector (RAD):**
The specific device is called the Radiation Assessment Detector which its primary focus is to detect radiation in order to determine shielding needs for potential human travelers, it has the ability to detect p, He, ions(Li-O), ions(Mg-Fe), neutrons, gamma rays, electrons, and positrons[8]. For gamma rays the radiation assessment detector has the ability to detect wavelength up to 7.9956×10^{-4} nm, for neutrons it has the ability to detect a wavelength of 1.199×10^4 nm, and for electrons it has the ability to detect 7.99×10^{-3} nm. This was determined by using the Planck's equation $E=h*c$.
- **Data Acquisition System: Slice6 Air:**

The radiation assessment detector will need to be compatible with a data acquisition system, the function of the data acquisition system is to store the data collected until it is sent back to Earth through the antenna. The data acquisition unit chosen is the Slice 6Air.

6.6.1.2. Physical characteristics

Table 6.5. Science Components

Components	Mass (Kg)	Dimensions	Power (W)
Radiation Assessment Detector	1.5 kg	10.3 x 12.2 x 20.4 cm	4.2 W
Slice6 Air	0.05 kg	4.2 x 4.2 x 1.3 cm	3.3 W

6.6.2. Design

6.6.2.1. Design approach

The Science Sub-Subsystem was decided on from the start of the Avionics subsystem as a means of organizing the subsystem into smaller sections for better organization.

6.6.2.2. Analysis tools

SolidWorks was used to design components and determine their placing to ensure that all science components are shielded from radiation and kept within their operating temperature range.

6.7. Communication

6.7.1. Characteristics

Communications handles all devices for receiving and sending data to Earth and customers.

6.7.1.1. Performance

- **Antenna: Oxford Space Systems Hinged Rib Antenna**
The Deployable Hinged Rib Antenna will communicate in the Ku band which has a bandwidth of 12-18 GHz and will send/receive data in the Ka band which has a bandwidth of 27-40 GHz.
- **Traveling Wave Tube Amplifier: CPI TWTA**
The Traveling wave tube amplifier used with the antenna will amplify the signals during communication across a frequency range of 18 to 26.5 GHz.
- **JPL/SDL Radio: IRIS V2**
Another component vital to sending/receiving data is the JPL/SDL Radio which is compatible with our Antenna, the JPL/SDL Radio uses the Ka band.

6.7.1.2. Physical characteristics

Table 6.6. Communication Components

Components	Mass (kg)	Dimensions	Power
Oxford Space Systems Hinged Rib Antenna	3 kg	Non-Deployed: 60 cm Deployed: 3 m	

CPI Traveling Wave Tube Amplifier	22.2kg	24.4 x 30 x 52.2 cm	45 W
IRIS V2 Radio	1.2 kg	10 x 10 x 56 cm	35 W

■

6.7.2. Design

6.7.2.1. Design approach

The Communication Sub-Subsystem was decided on from the start of the Avionics subsystem as a means of organizing the subsystem into smaller sections for better organization.

6.7.2.2. Analysis tools

SolidWorks was used to determine the placing of the antenna to ensure that no other components are interfering with the antenna's field of view.

6.8. Power

6.8.1. Characteristics

The power sub-subsystem oversees how much power is used by all components, and how much power will be generated by the solar panels.

6.8.1.1. Performance

- Solar Panels and Rechargeable Batteries

It was decided from early requirements that our platform would function using Nickel -Hydrogen rechargeable batteries and solar panels

6.8.1.2. Physical characteristics

Table 6.7. Power Budget

Power Budget					
Estimated Automated Refueling Cislunar Outpost (ARCO) Power Budget					
Subsystems	Operation	Components	Duty Cycle (%)	Power Consumption (W)	Quantity
Avionics:					
	On-Board Computer Handling:				
		On-Board Computer	100%	15	1
	Data Handling:				
		Data Acquisition System	25%	3.3	1
	Telecommunication Equipment:				
		JPL/SDL Radio	50%	35	1
		Antenna/TWTA	100%	45	1
GNC:					
	Attitude Control Equipment				
		Star Trackers	100%	1.54	2
		Reaction Wheels	100%	90	3
Thermal System					
	Thermal Control:				
		Heaters	100%	300	1
Propulsion:					
	Propulsion Thruster Valves:				
		Biopropellant TR-500 SCAT		140.6	4
		Monopropellant MRE-1		15	8
Payload:					
	Radiation Detection:				
		Radiation Assesment Detector	25%	4.2	1
	Refueling Docking:				
		RAFTI	4.20%	10	1
Platform Subtotal:				1367.98	
Margin				10%	
Total:				1504.778	

6.8.2. Design

6.8.2.1. Design approach

The Power Sub-Subsystem was decided on from the start of the Avionics subsystem as a means of organizing the subsystem into smaller sections for better organization.

6.8.2.2. Analysis tools

SolidWorks was used to determine the placing and design of solar panels, Matlab was used to run all calculations for the sizing of solar panels. Excel Spreadsheet was used to calculate the total power consumption of ARCO.

7. Structures Subsystem

7.1. Characteristics

7.1.1. Performance

The structures subsystem is vital to all other subsystem components and payloads. The structural frame of the spacecraft will ensure that all over components are protected from the harsh environment of space and launch as well as act as a mounting surface for critical electronics. The structures subsystem is driven by a specific set of subsystem level requirements that are derived from higher level customer and systems requirements:

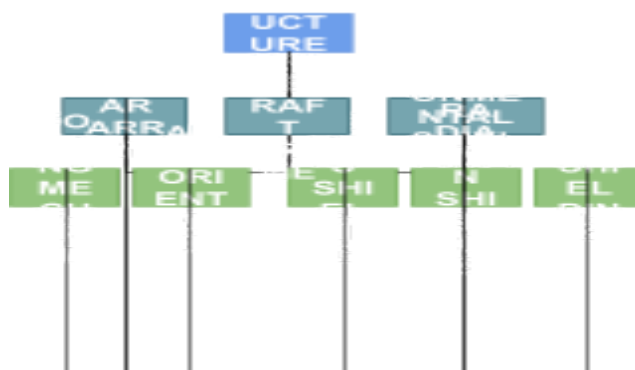
Subsys Req ID	Structures Subsystem Requirement
---------------	----------------------------------

SSR 1.1.1	Subsystem chassis shall be constructed using 6061 Aluminum alloy for optimal strength and mass characteristics
SSR 1.1.2	Subsystem chassis shall utilize a series of welded beams in an octagon shape for optimal mass characteristics
SSR 1.2.1	Subsystem shall utilize deployment mechanisms for folding/unfolding communications equipment
SSR 1.2.2	Subsystem shall utilize a hinged solar array
SSR 1.3.1	Subsystem shall withstand 1000 kN of force during launch
SSR 1.4.1	Subsystem shall utilize Gallium Arsenide solar cells
SSR 1.5.1	Subsystem shall utilize an 2024 Aluminum whipple shield to protect against space debris
SSR 1.5.2	Subsystem shall feature a 45 mm thick stuffed layer of Kevlar to further protect against space debris
SSR 1.6.1	Subsystem shall utilize a 1 mm thick layer of high density Polyethylene panels to protect against ionizing radiation

Table 7.1. Structures Subsystem Requirements

7.1.2. External Interfaces

As shown in Figure 7.1, the structures subsystem follows a hierarchy of component development that flows down into specific subcomponents. Each subcomponent fulfills a critical mission need for our spacecraft.



7.1.3. Physical Characteristics

From the subsystem requirements, solutions have been developed in accordance with the specific mission goals. The spacecraft frame was modeled as an aluminum octagonal cylinder with a diameter of 3.355 meters and a height of 3 meters. Aluminum 6061 was chosen because of its high strength value compared to its mass.

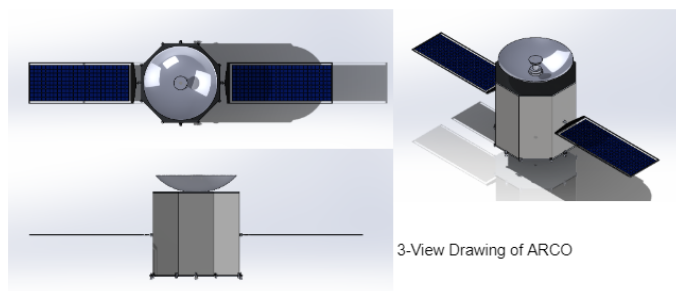


Figure 7.2: Model of ARCO

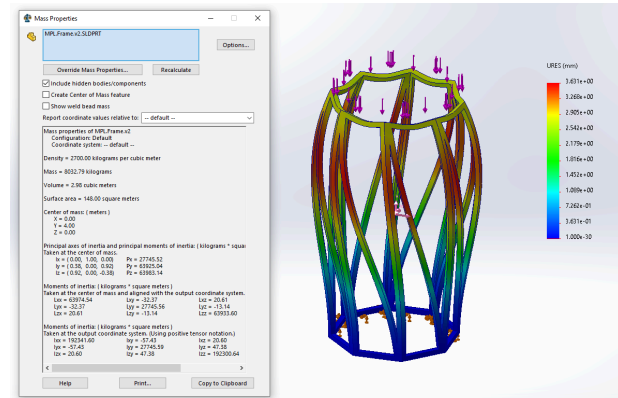
7.2. Design

7.2.1. Design Approach

The structures subsystem was designed in regards to the figures laid out in the system level and subsystem level requirements. When starting the design process for the three structures components, it was paramount that the subsystem fit within the mass and volume constraints given by the SpaceX Falcon Heavy Payload User Guide. Following the PUG, the spacecraft must not exceed a mass of 16800 kg. Additionally, the spacecraft must fit in the payload fairing, which has a diameter of 180 inches (4.572 meters) and a height of 453 inches (11.5062 meters).

7.2.2. Analysis Tools

As the design process matures, analysis of each component is done to further optimize the subsystem. For the frame this optimization was primarily done through the use of the SolidWorks simulations feature coupled with the mass calculations function shown.



7.2.3. Trade Studies

Before developing the design, trade studies were conducted in order to gather more information on potential materials and processes for the structures subsystem.

For the shield design, multiple different materials were researched and compared for strength and density properties. As shown below, these materials were tabulated in order to see the most viable options. After comparison, it was decided to move forward with Aluminium, Kevlar, and Polyethylene for the Whipple Shield

Structural Material	Type	Strength	Density	Cost
Aluminum 2024 Panel	Panel	186 MPa	2.78 g/cm3	\$64.67/kg
Aluminum Mesh	Fabric	n/a	2.7 g/cm3	\$19/m2
Kevlar	Fabric	3620 MPa	1.4 g/cm3	\$50/kg
Polyethylene	Fabric	30 Mpa	0.95 g/cm3	\$11/m2
Molybdenum	Panel	324 MPa	10.2 g/cm3	\$85/kg
Tungsten Carbide	Coating	344 MPa	15.63 g/cm3	\$142.60/kg
Oxidized Metal	Coating	n/a	5.242 g/cm3	\$50.6/kg

[7.1], [7.2], [7.3]

Table 7.2. Structure Materials

7.3. Construction

For the structures subsystem, both manufacturing and integration will be needed in tandem in order to fulfill the requirements of this section

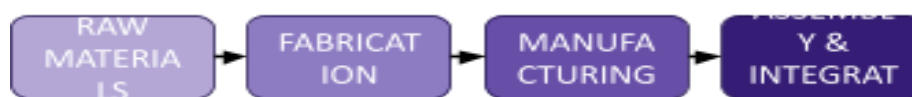


Figure 7.4: Flowchart of Manufacturing and Integration Process

In regards to the shield and frame designs, the construction process will follow a simple process in which a predetermined amount of raw material is procured and is then fabricated into the required panels and beams. From there, the individual components (shield and frame) are manufactured using a combination of welding and fastening. Finally these components are assembled and integrated with the solar panels to form the structures subsystem.

Structures Subsystem Cost Breakdown	
Solar Array	\$37,500,000
Spacecraft Frame	\$10,000
Spacecraft Shield	\$150,000
Total	\$37,660,000

Table 7.3. Structures Costs

7.4. Verification

To verify that the subsystem fulfills the systems requirements, further testing and analysis will need to be conducted. Additionally more in depth calculations like the one shown below will need to be completed to ensure the subsystem is within mass and cost requirements.

Material	Mass (kg)	Price (per kg)	Cost (\$)
Rear Wall Al	237.6260095	2.31	548.916082
Poly Layer	27.08814554	1.18	31.96401174
Kevlar	1821.849637	70	127529.4746
Bumper Al	81.43763097	2.31	188.1209275
	Total Mass (kg)		Estimated Cost (\$)
	2168.001423		128298.4756

[7.4], [7.5]

Table 7.4. Structures Masses

Similarly, simulations will also be used as part of the frame verification to further understand how the component will handle launch conditions. Using both Static Load and Random Vibration conditions to demonstrate the viability of the component.

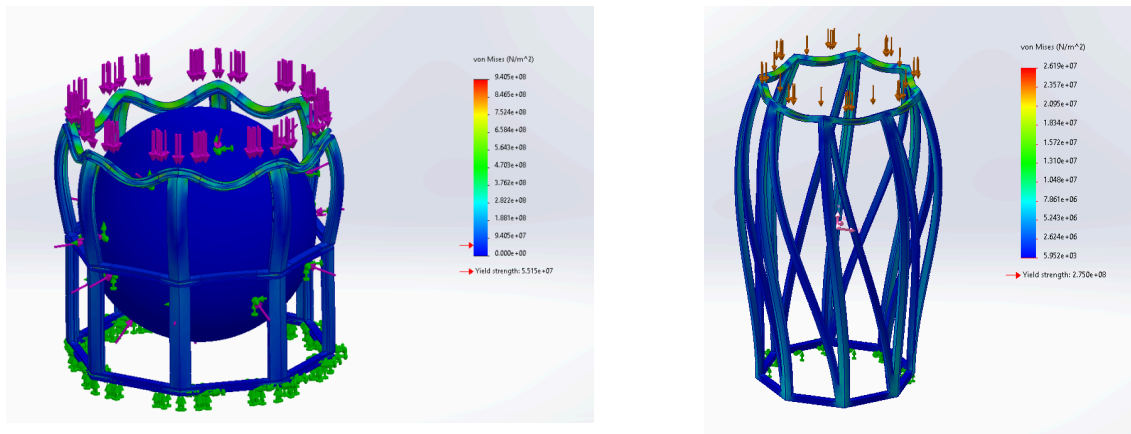


Figure 7.5. a & b: Analysis on Frame Structure

Structures Subsystem Verification	
Solar Array	Inspection, Demonstration
Spacecraft Frame	Test, and Analysis
Spacecraft Shield	Demonstration, Test, and Analysis

Table 7.5. Structures Verification

7.5. Structures Components

7.5.1. Solar Array:

The Solar Array is vital for the collection of solar power and the spacecraft's continued operation. The design features a stowed and deployed configuration exhibiting the "Roll Out Solar Array" devices developed by Redwire Space.

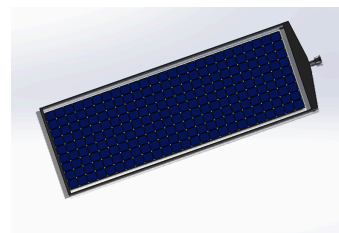


Figure 7.6: Model of Solar Array

The "Roll Out Solar Array" of this spacecraft features 2 modules with the capability of providing 1691.72 W of power.

7.5.2. Spacecraft Frame:

Additionally, the spacecraft frame ensures that all vital electronics and payloads are adequately protected during launch. The required amount of force the structure will need to withstand according to the Falcon Heavy user guide is expected to be around 1000 KN.

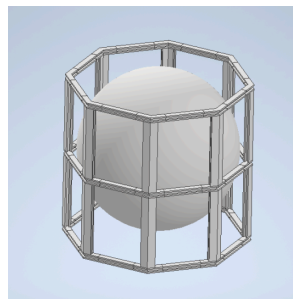


Figure 7.7: Model of Spacecraft Frame

A truss design was applied to the frame in order to further support joints that would be under significant loading during launch. This design was further optimized by minimizing the cross sectional area of the aluminum beams.

7.5.3. Spacecraft Shield:

Finally, the design will need a dedicated shielding component in order to protect spacecraft vitals such as the fuel and oxidizer tanks from the unstable L1 environment. The outer shield serves to negate the effects of ionizing radiation that would disrupt avionics and deflect micro-debris that would significantly damage the propellant tanks.

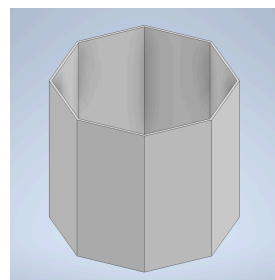


Figure 7.8: Model of Shield

This shield features a 3 millimeter Aluminum 2024 shield followed by a 1 millimeter layer of Polyethylene. This is then wrapped in a 45 millimeter layer of Kevlar. Finally the outer layer is covered in 1 millimeter Aluminum panels.

8. Docking Subsystem

8.1. Characteristics

8.1.1. Performance

The docking and refueling system should be able to perform 50 cycles. The RAFTI system meets the cycle life requirement because it has a life cycle of 1,000 cycles^[8.2]. The RAFTI system is able to dock with other spacecraft with incoming velocities of up to 25 mm/s^[8.3]. The RAFTI system is able to refuel other spacecraft at a rate of 4 L/min^[8.3]. With RAFTI as our docking and refueling port, ARCO will be able to refuel another spacecraft with 225 kg of hydrazine propellant, in just under an hour. The docking will occur with the help of a clamp that the customer will have. The spacecraft will have at least 3 fiducials that will be used to communicate with the customer spacecraft to do a cooperative dock^[8.3].

Table 8.1. Docking Subsystem Requirements

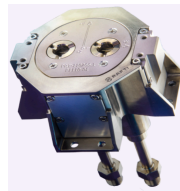
Subsystem Requirement ID	Docking Subsystem Requirement
SSR 2.1.1	Docking system shall be able to cycle at least 50 times.
SSR 2.1.2	Docking port shall refuel at a rate of at least 4 L/min.
SSR 2.2.1	Docking port shall support a mass of at least 16,800 kg.
SSR 2.2.2	Docking system shall dock at a velocity less than 30 mm/s.

8.1.2. External Interface Characteristics

In order for the refueling and docking system to be automated, there will be at least 3-4 fiducials^[8], which are referencing points, used to guide the customer spacecraft to the docking. Without these fiducials it will not be possible to dock with spacecraft autonomously. These 3-4 fiducials will be placed 90 degrees away from each other and 200 mm away from the center of the RAFTI port^[8.3].

8.1.3. Physical Characteristics

Table 8.2: RAFTI Physical Characteristics^[8.3]

Component	Mass (kg)	Dimensions (mm)	Power	Image
RAFTI	0.5 kg	45 x 75 x 75 (External part)	10 W	

8.1.4. Other Characteristics

Table 8.3: RAFTI Characteristics^[8.3]

Component	Survival Temperatures	Operating Temperatures	Cycle Life	Pressure	Flow Rate
RAFTI	-60°C to 100°C	-31°C to 70°C	1000 cycles	15 psig	4 L/min

8.2. Design

Compared to PDR we decided that the SPHERES universal docking port was unnecessary since the RAFTI system was able to support a docking mass of 16,800 kg. This meant that the SPHERES universal docking port was going to have extra mass, extra cost, and extra power usage that was not going to be needed. Also compared to PDR the refuelant tank was changed from a cylindrical type of tank into a spherical type of tank as it will best fit the spacecraft.

8.2.1. Design Approach

Figure 8.1.: Docking Subsystem Flow



8.2.2. Analysis Tool

Solidworks was used to determine where it was best to place the RAFTI system since it needs a 520 mm keepout zone so that no other component would be in the way of the keepout zone. Also solidworks was used to find the mass of the refueling propellant tank.

8.2.3. Trade Studies and Selection

For the docking subsystem, we had to choose between the commercial off the shelf universal docking ports in the industry. Most universal docking ports were for human spaceflight, so that narrowed our search to the SPHERES docking port. We were faced with the same challenge for the refueling port of finding a non-human spaceflight refueling port, and that led us to the RAFTI universal refueling port. After finding more information on the RAFTI system, it was determined that the SPHERES universal docking was not needed since the RAFTI system meets with the requirements. For our propellant tank sizing, we have used the most common type of propellant tank on the market, a thin-walled aluminum tank, and sized it to fit inside the shell of our spacecraft. The selection of the tank was determined by the amount of fuel needed to refuel 50 spacecraft over a 10 year period and so it was between choosing a cylindrical or spherical tank. The Spherical tank was chosen because that would best fit and carry the enough amount of fuel needed for the mission. Even though a cylindrical tank would have a higher volume it would increase the dry mass more than a spherical tank which would not be needed since the spherical tank size is less weight and can carry the amount of fuel needed to fuel 50 spacecrafts which is a requirement.

8.3. Construction

RAFTI will be attached through a small mounting board on the zenith-facing side of the spacecraft in such a way that it will stay aligned with the center of mass. There are two fluid interfaces that run through RAFTI, one for Hydrazine propellant and the other for Helium pressurant. These fluid interfaces will run piping directly to the two separate tanks, the pressurant feeding the propellant to the customer. The Hydrazine tank will be spherical and have thin baffles placed intermittently to reduce propellant slosh. The tanks will both be secured to the frame by bulkheads and mounting surfaces at the fluid exit points.

8.4. Verification

Three major verifications of the docking subsystem need to take place. First, the temperatures experienced inside the tank must be shown not to dip or rise above the operating temperatures of Hydrazine. Next, the tank thicknesses need to be verified to ensure that there is no rupture due to pressure overload. Finally, the mass of each part of the docking system flow needs to be validated to ensure ARCO is under the required launch mass.

Table 8.4: Docking Subsystem Verification

Tank thermal management	Solidworks simulation of the spacecraft's internal thermal control to ensure Hydrazine's operating temperatures are not exceeded.
Propellant and Pressurant tank thickness	Solidworks simulation of the tanks under the maximum pressure loading.
Mass	Solidworks evaluation of the mass properties of the docking port and propellant tanks.

9. Appendices

A. Individual contribution

Alex Uribe:

- Conducted research on materials and structural designs
- Led spacecraft shield development
- Assisted in modeling components in CAD and integrating assembly
- Primarily wrote section 7

Samuel Estrada:

- Researched on the docking and refueling ports
- modeled the RAFTI system in CAD
- Helped write executive summary

Summer Huang:

- everything of propulsion

Elizabeth Cotto:

- Researched different components that would best satisfy functions (radiation detector, antenna, JPL Radio, Batteries, Solar Panels, etc.)
- Approximated mass of subsystem components
- Approximated sizing of panels
- Assisted in Trade studies conducted for the Avionics Subsystem.

Angelica Cortez:

- Researched components that could be used at L1 for the avionics subsystem
- helped with research done for solar array design and mass
- conducted trade studies for why certain components were chosen
- contacted multiple companies to find mass and cost of components for ARCO

Andrew Mills:

- Led project tasking
- Docking subsystem member
- Wrote/helped with sections 1-3,8
- Helped write executive summary

Andrea Coca Gonzales :

- Worked on Simulink verification for star trackers
- Helped sizing wheels and calculating external perturbances at L1
- Researched recalibration methods for star trackers
- Code for reaction wheel sizing
- GNC star tracker section

Juan Flores:

- Researched reaction wheels
- helped with research on star trackers
- Code for reaction wheel sizing
- GNC reaction wheel section + Customer/System Requirements

Ernest Franco:

- Researched solar array design
- Assembled system integration CAD on solidworks
- Helped with sections 3 and 7

- Optimized system from PDR to account for mass requirement

B. References

Applicable Documents:

- [2.1] *Evolved expendable launch vehicle rideshare user's guide* - NASA. (n.d.). Retrieved March 22, 2023, from [https://explorers.larc.nasa.gov/HPSMEX/SMEX/pdf_files/EELV-RideshareUsersGuide-v1-Signed\[6\].pdf](https://explorers.larc.nasa.gov/HPSMEX/SMEX/pdf_files/EELV-RideshareUsersGuide-v1-Signed[6].pdf)
- [2.2] NASA. (n.d.). *NASA Technical Standards*. NASA. Retrieved March 21, 2023, from <https://standards.nasa.gov/nasa-technical-standards>
- [2.3] *Space mission engineering: The new smad*. Space Mission Engineering: The New SMAD. (n.d.). Retrieved March 21, 2023, from <http://www.sme-smad.com/>

Propulsion:

- [4.1] Northrop Grumman (n.d.). *PROPULSION PRODUCTS AND SERVICES*. PROPULSION PRODUCTS AND SERVICES. <https://www.northropgrumman.com/space/propulsion-products-and-services/>
- [4.2] (n.d.). *Titanium Bipropellant Tanks for Satellites and Spacecraft*. Ariane Group. <https://www.space-propulsion.com/spacecraft-propulsion/bipropellant-tanks/index.html#2100>
- [4.3] (n.d.). *Valves for Spacecraft Propulsion Systems*. Moog. Retrieved March 21, 2023, from <https://www.moog.com/products/propulsion-controls/spacecraft/spacecraft-propulsion-components/valves-for-spacecraft-propulsion-systems.html>
- [4.4] (n.d.). *EARTH-MOON LIBRATION STATIONKEEPING: THEORY, MODELING, AND OPERATIONS*. Ucla.edu. https://artemis.igpp.ucla.edu/pubs/2012_nonrefereed/Folta_ IAA_12.pdf
- [4.5] (n.d.). *James Webb Space Telescope Initial Mid-Course Correction Monte Carlo Implementation using Task Parallelism*. ISSFD. https://issfd.org/ISSFD_2014/ISSFD24_Paper_S13-2_Petersen.pdf
- [4.6] (n.d.). *Trajectory Design with STK/Astrogator*. Astrogatorsguild.com. https://astrogatorsguild.com/wp-content/Astrogator_Training/EM_L1_Tutorial.pdf

GNC:

- [5.1] "RSI 18-220/45." *Satsearch*, <https://satsearch.co/products/collins-aero-rsi-18-220-45>.
- [5.2] "Star Trackers." Rocket Lab, <https://www.rocketlabusa.com/space-systems/satellite-components/star-trackers/>
- [5.3] "Tiny Star Trackers Help Spacecraft Find Their Place." NASA, NASA, https://spinoff.nasa.gov/Spinoff2019/it_2.html#:~:text=The%20star%20tracker%20uses%20the, person%20to%20do%20the%20calculation
- [5.4] "Three-Axis Configuration." *Google Search*, Google, https://www.google.com/search?q=reaction%2Bwheel%2Bthree%2Baxis&rlz=1C1CHBF_enUS904US904&sxsrf=AJOqlzUZD7GRNJ3EW0PMB7y_qYS9qOHucQ%3A1679461671354&source=lnms&tbm=isch&sa=X&ved=2ahUKEwiA2dai4u79AhVaMUQIHcCFD8kQ0pQJegQIAxAC&biw=1620&bih=937&dpr=2#imgsrc=4_X2yRSkLq0uNM
- [5.5] Brown, Charles D. *Elements of Spacecraft Design*. American Institute of Aeronautics and Astronautics, 2002.

Avionics:

- [6.1] SuperManu. "Space Exploration." *Stack Exchange*, Stack Exchange, 11 Mar. 2020, <https://space.stackexchange.com/questions/41769/for-how-long-do-the-various-earth-moon-lagrangian-points-receive-sunlight-each-m>.
- [6.2] Brown, Charles D. *Elements of Spacecraft Design*, AIAA, Reston, Va, 2002.
- [6.3] Caldwell, Sonja. "9.0 Communications." NASA, 16 Oct. 2021, www.nasa.gov/smallsat-institute/sst-soa/communications#9.5.7.
- [6.4] Zhu, Frances. "5.5 Power Generation." *A Guide to CubeSat Mission and Bus Design*, <https://pressbooks-dev.oer.hawaii.edu/epet302/chapter/5-5-power-generation/>.
- [6.5] Hassler, D. et.al. "The Radiation Assessment Detector (RAD) Investigation." *SCISPACE*. <https://typeset.io/papers/the-radiation-assessment-detector-rad-investigation-4aywwq29pk>

Structures:

- [7.1] World Material. (n.d.). *Weight & density of aluminum 6061 g/cm3, lbs/IN3, KG/m3, g/ml, LB/FT3, G/MM3, cubic inch*. World Material. Retrieved March 21, 2023, from <https://www.theworldmaterial.com/weight-density-of-aluminum/>
- [7.2] Kevlar: Properties, Price & Application. Material Properties. (2021, June 26). Retrieved March 21, 2023, from <https://material-properties.org/kevlar-properties-application-price/>
- [7.3] Dielectric Manufacturing. (2022, July 25). *Material properties of polyethylene (PE) thermoplastic - polymer*. Dielectric Manufacturing. Retrieved March 21, 2023, from <https://dielectricmfg.com/resources/knowledge-base/polyethylene/>
- [7.4] Aluminum price (I:APNM). Aluminum Price. (n.d.). Retrieved March 21, 2023, from https://ycharts.com/indicators/aluminum_price
- [7.5] High density polyethylene (HDPE) sheet. High Density Polyethylene (HDPE) Sheet | U.S. Plastic Corp. (n.d.). Retrieved March 21, 2023, from <https://www.usplastic.com/catalog/item.aspx?itemid=23869>

Docking:

- [8.1] Bultitude, James. "First Flight of RAFTI Orbit Refueling Interface." https://www.researchgate.net/Publication/356289451_First_Flight_of_RAFTI_Orbital_Refueling_Interface. 29 Oct. 2021.
- [8.2] "Products." *Orbit Fab | Gas Stations in Space™*. www.orbitfab.com/products.
- [8.3] RAFTI User Guide. www.orbitfab.com/products.